

Запрос	План на 10000 строк	План на 10000000 строк (без индексов)	Оптимизация (Что улучшено)	План на 10000000 строк (с индексами)
Выбор всех фильмов на сегодня <pre>SELECT hall_id, movie_id FROM screenings WHERE DATE(starts_at) = CURRENT_DATE;</pre>	"Seq Scan on screenings (cost=0.00..33.40 rows=3 width=16) (actual time=0.077..0.091 rows=8 loops=1)" " Filter: ((starts_at >= CURRENT_DATE) AND (starts_at < (CURRENT_DATE + '1 day'::interval))))" " Rows Removed by Filter: 32" "Planning Time: 0.203 ms" "Execution Time: 0.115 ms"	"Seq Scan on screenings (cost=0.00..1313.00 rows=1 width=16) (actual time=0.256..105.977 rows=6 loops=1)" " Filter: ((starts_at >= CURRENT_DATE) AND (starts_at < (CURRENT_DATE + '1 day'::interval))))" " Rows Removed by Filter: 39994" "Planning Time: 7.534 ms" "Execution Time: 106.001 ms"	Индекс CREATE INDEX idx_screenings_starts_at ON screenings(starts_at); Исправлен запрос, убрана функция DATE <pre>SELECT hall_id, movie_id FROM screenings WHERE starts_at >= CURRENT_DATE AND starts_at < CURRENT_DATE + interval '1 day'</pre>	"Index Scan using idx_screenings_starts_at on screenings (cost=0.30..8.40 rows=5 width=16) (actual time=0.100..0.103 rows=6 loops=1)" " Index Cond: ((starts_at >= CURRENT_DATE) AND (starts_at < (CURRENT_DATE + '1 day'::interval))))" "Planning Time: 4.214 ms" "Execution Time: 0.147 ms"
Подсчёт проданных билетов за неделю <pre>SELECT COUNT(*) FROM tickets WHERE status IN ('paid', 'reserved') AND screening_id IN (SELECT id FROM screenings WHERE starts_at >= NOW() - INTERVAL '7 days');</pre>	"Aggregate (cost=307.85..307.86 rows=1 width=8) (actual time=12.452..12.456 rows=1 loops=1)" " -> Hash Join (cost=41.48..282.85 rows=10000 width=0) (actual time=6.753..11.880 rows=10000 loops=1)" " Hash Cond: (tickets.screening_id = screenings.id)" " -> Index Only Scan using	"Finalize Aggregate (cost=158170.59..158170.60 rows=1 width=8) (actual time=2100.783..2122.514 rows=1 loops=1)" " -> Gather (cost=158170.37..158170.58 rows=2 width=8) (actual time=2098.438..2122.395 rows=3 loops=1)" " Workers Planned: 2" " Workers Launched: 2" " -> Partial Aggregate (cost=157170.37..157170.38 rows=1 width=8) (actual time=2011.893..2011.896 rows=1 loops=3)"	Индекс CREATE INDEX idx_tickets_screening_status ON tickets (screening_id, status); Оптимизирован запрос. Сделал явный JOIN <pre>SELECT COUNT(*) FROM tickets t JOIN screenings s ON t.screening_id = s.id WHERE t.status IN ('paid', 'reserved') AND s.starts_at >= NOW() - INTERVAL '7 days';</pre>	"Finalize Aggregate (cost=158172.31..158172.32 rows=1 width=8) (actual time=1889.532..1900.516 rows=1 loops=1)" " -> Gather (cost=158172.09..158172.30 rows=2 width=8) (actual time=1887.936..1900.481 rows=3 loops=1)" " Workers Planned: 2" " Workers Launched: 2" " -> Partial Aggregate (cost=157172.09..157172.10 rows=1 width=8) (actual time=1813.080..1813.082 rows=1 loops=3)"

	idx_tickets_screening_status on tickets (cost=0.29..215.28 rows=10000 width=8) (actual time=6.644..9.935 rows=10000 loops=1)" " Filter: ((status)::text = ANY ('{paid,reserved}'::text[]))" " Heap Fetches: 0" " -> Hash (cost=28.20..28.20 rows=1040 width=8) (actual time=0.051..0.053 rows=40 loops=1)" " Buckets: 2048 Batches: 1 Memory Usage: 18kB" " -> Seq Scan on screenings (cost=0.00..28.20 rows=1040 width=8) (actual time=0.019..0.030 rows=40 loops=1)" " Filter: (starts_at >= (now() - '7 days'::interval))" "Planning Time: 41.784 ms" "Execution Time: 12.593 ms"	" -> Hash Join (cost=1613.38..146754.80 rows=4166229 width=0) (actual time=50.062..1749.769 rows=3333333 loops=3)" " Hash Cond: (tickets.screening_id = screenings.id)" " -> Parallel Index Only Scan using idx_tickets_screening_status on tickets (cost=0.43..134203.25 rows=4166645 width=8) (actual time=19.289..883.470 rows=3333333 loops=3)" " Filter: ((status)::text = ANY ('{paid,reserved}'::text[]))" " Heap Fetches: 0" " -> Hash (cost=1113.00..1113.00 rows=39996 width=8) (actual time=29.698..29.698 rows=40000 loops=3)" " Buckets: 65536 Batches: 1 Memory Usage: 2075kB" " -> Seq Scan on screenings (cost=0.00..1113.00 rows=39996 width=8) (actual time=0.566..16.578 rows=40000 loops=3)" " Filter: (starts_at >= (now() - '7		" -> Hash Join (cost=1613.43..146755.43 rows=4166667 width=0) (actual time=48.395..1574.938 rows=3333333 loops=3)" " Hash Cond: (t.screening_id = s.id)" " -> Parallel Index Only Scan using idx_tickets_screening_status on tickets t (cost=0.43..134203.77 rows=4166667 width=8) (actual time=14.474..778.748 rows=3333333 loops=3)" " Filter: ((status)::text = ANY ('{paid,reserved}'::text[]))" " Heap Fetches: 0" " -> Hash (cost=1113.00..1113.00 rows=40000 width=8) (actual time=32.342..32.343 rows=40000 loops=3)" " Buckets: 65536 Batches: 1 Memory Usage: 2075kB" " -> Seq Scan on screenings s (cost=0.00..1113.00 rows=40000 width=8) (actual time=0.595..18.318 rows=40000 loops=3)" " Filter: (starts_at >= (now() - '7 days'::interval))" "Planning Time: 6.993 ms"
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		days'::interval)))" "Planning Time: 0.603 ms" "JIT:" " Functions: 38" " Options: Inlining false, Optimization false, Expressions true, Deforming true" " Timing: Generation 10.130 ms, Inlining 0.000 ms, Optimization 3.128 ms, Emission 52.150 ms, Total 65.408 ms" "Execution Time: 2124.736 ms"		"JIT:" " Functions: 38" " Options: Inlining false, Optimization false, Expressions true, Deforming true" " Timing: Generation 5.713 ms, Inlining 0.000 ms, Optimization 1.750 ms, Emission 40.477 ms, Total 47.940 ms" "Execution Time: 1902.679 ms"
Формирование афиши (фильмы, которые показывают сегодня) SELECT m.title, s.starts_at, c.name AS cinema, h.name AS hall FROM screenings s JOIN movies m ON m.id = s.movie_id JOIN halls h ON h.id = s.hall_id JOIN cinemas c ON c.id = h.cinema_id WHERE DATE(s.starts_at) = CURRENT_DATE ORDER BY s.starts_at;	"Sort (cost=46.19..46.21 rows=5 width=660) (actual time=0.940..0.943 rows=8 loops=1)" " Sort Key: s.starts_at" " Sort Method: quicksort Memory: 25kB" " -> Nested Loop (cost=28.85..46.14 rows=5 width=660) (actual time=0.842..0.863 rows=8 loops=1)" " -> Hash Join (cost=28.71..44.57 rows=5 width=250) (actual time=0.377..0.384 rows=8 loops=1)" " Hash Cond: (h.id =	"Sort (cost=1146.56..1147.06 rows=200 width=660) (actual time=7.935..7.941 rows=6 loops=1)" " Sort Key: s.starts_at" " Sort Method: quicksort Memory: 25kB" " -> Hash Join (cost=18.01..1138.91 rows=200 width=660) (actual time=0.876..7.858 rows=6 loops=1)" " Hash Cond: (h.cinema_id = c.id)" " -> Nested Loop (cost=6.66..1127.03 rows=200 width=250) (actual time=0.625..7.604 rows=6 loops=1)" " -> Hash Join (cost=6.50..1120.04 rows=200	Индекс CREATE INDEX idx_screenings_movie_id_st arts_at ON screenings(movie_id, starts_at); Исправил запрос, убрал DATE, чтобы индекс сработал SELECT m.title, s.starts_at, c.name AS cinema, h.name AS hall FROM screenings s JOIN movies m ON m.id = s.movie_id JOIN halls h ON h.id = s.hall_id JOIN cinemas c ON c.id =	"Sort (cost=31.81..31.82 rows=5 width=660) (actual time=0.271..0.274 rows=6 loops=1)" " Sort Key: s.starts_at" " Sort Method: quicksort Memory: 25kB" " -> Nested Loop (cost=14.46..31.75 rows=5 width=660) (actual time=0.226..0.239 rows=6 loops=1)" " -> Hash Join (cost=14.32..30.19 rows=5 width=250) (actual time=0.193..0.198 rows=6 loops=1)" " Hash Cond: (h.id = s.hall_id)" " -> Seq Scan on halls h (cost=0.00..13.10 rows=310

s.hall_id)" " -> Seq Scan on halls h (cost=0.00..13.10 rows=310 width=234) (actual time=0.248..0.249 rows=4 loops=1)" " -> Hash (cost=28.64..28.64 rows=5 width=32) (actual time=0.106..0.107 rows=8 loops=1)" " Buckets: 1024 Batches: 1 Memory Usage: 9kB" " -> Merge Join (cost=28.49..28.64 rows=5 width=32) (actual time=0.080..0.086 rows=8 loops=1)" " Merge Cond: (m.id = s.movie_id)" " -> Index Scan using movies_pkey on movies m (cost=0.14..16.14 rows=200 width=24) (actual time=0.022..0.026 rows=15 loops=1)" " -> Sort (cost=28.26..28.27 rows=5 width=24) (actual time=0.048..0.049 rows=8 loops=1)" " Sort Key: s.movie_id"	width=32) (actual time=0.168..7.132 rows=6 loops=1)" " Hash Cond: (s.movie_id = m.id)" " -> Seq Scan on screenings s (cost=0.00..1113.00 rows=200 width=24) (actual time=0.013..6.970 rows=6 loops=1)" " Filter: (date(starts_at) = CURRENT_DATE)" " Rows Removed by Filter: 39994" " -> Hash (cost=4.00..4.00 rows=200 width=24) (actual time=0.136..0.138 rows=200 loops=1)" " Buckets: 1024 Batches: 1 Memory Usage: 19kB" " -> Seq Scan on movies m (cost=0.00..4.00 rows=200 width=24) (actual time=0.005..0.028 rows=200 loops=1)" " -> Memoize (cost=0.16..0.42 rows=1 width=234) (actual time=0.077..0.077 rows=1 loops=6)" " Cache Key: s.hall_id"	h.cinema_id WHERE s.starts_at >= CURRENT_DATE AND s.starts_at < CURRENT_DATE + interval '1 day' ORDER BY s.starts_at;	width=234) (actual time=0.016..0.017 rows=4 loops=1)" " -> Hash (cost=14.26..14.26 rows=5 width=32) (actual time=0.148..0.149 rows=6 loops=1)" " Buckets: 1024 Batches: 1 Memory Usage: 9kB" " -> Hash Join (cost=8.46..14.26 rows=5 width=32) (actual time=0.103..0.137 rows=6 loops=1)" " Hash Cond: (m.id = s.movie_id)" " -> Seq Scan on movies m (cost=0.00..4.00 rows=200 width=24) (actual time=0.009..0.024 rows=200 loops=1)" " -> Hash (cost=8.40..8.40 rows=5 width=24) (actual time=0.077..0.077 rows=6 loops=1)" " Buckets: 1024 Batches: 1 Memory Usage: 9kB" " -> Index Scan using idx_screenings_starts_at on screenings s (cost=0.30..8.40 rows=5 width=24) (actual time=0.054..0.058 rows=6 loops=1)"
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	Sort Method: quicksort Memory: 25kB" " -> Seq Scan on screenings s (cost=0.00..28.20 rows=5 width=24) (actual time=0.013..0.018 rows=8 loops=1)" Filter: (date(starts_at) = CURRENT_DATE)" Rows Removed by Filter: 32" " -> Index Scan using cinemas_pkey on cinemas c (cost=0.14..0.31 rows=1 width=426) (actual time=0.059..0.059 rows=1 loops=8)" " Index Cond: (id = h.cinema_id)" "Planning Time: 44.510 ms" "Execution Time: 1.167 ms"	Cache Mode: logical" Hits: 2 Misses: 4 Evictions: 0 Overflows: 0 Memory Usage: 1kB" " -> Index Scan using halls_pkey on halls h (cost=0.15..0.41 rows=1 width=234) (actual time=0.113..0.113 rows=1 loops=4)" " Index Cond: (id = s.hall_id)" " -> Hash (cost=10.60..10.60 rows=60 width=426) (actual time=0.230..0.231 rows=2 loops=1)" " Buckets: 1024 Batches: 1 Memory Usage: 9kB" " -> Seq Scan on cinemas c (cost=0.00..10.60 rows=60 width=426) (actual time=0.211..0.212 rows=2 loops=1)" "Planning Time: 11.855 ms" "Execution Time: 8.163 ms"		Index Cond: ((starts_at >= CURRENT_DATE) AND (starts_at < (CURRENT_DATE + '1 day'::interval)))" " -> Index Scan using cinemas_pkey on cinemas c (cost=0.14..0.31 rows=1 width=426) (actual time=0.006..0.006 rows=1 loops=6)" " Index Cond: (id = h.cinema_id)" "Planning Time: 1.545 ms" "Execution Time: 0.497 ms"
Поиск 3 самых прибыльных фильмов за неделю SELECT m.title, SUM(t.price_amount) AS total	"Limit (cost=359.73..359.73 rows=3 width=48) (actual time=44.375..44.383 rows=3 loops=1)" " -> Sort (cost=359.73..360.23 rows=200 width=48) (actual time=44.373..44.379 rows=3	"Limit (cost=212974.13..212974.13 rows=3 width=48) (actual time=15624.124..15636.186 rows=3 loops=1)" " -> Sort (cost=212974.13..212974.63	Индекс CREATE INDEX idx_tickets_order_id_paid ON tickets (order_id, screening_id, price_amount) WHERE status = 'paid';	"Limit (cost=212974.67..212974.68 rows=3 width=48) (actual time=14786.142..14825.150 rows=3 loops=1)" " -> Sort (cost=212974.67..212975.17

FROM tickets t JOIN screenings s ON t.screening_id = s.id JOIN movies m ON s.movie_id = m.id JOIN orders o ON t.order_id = o.id WHERE t.status = 'paid' AND o.paid_at >= NOW() - INTERVAL '7 days' GROUP BY m.title ORDER BY total DESC LIMIT 3;	loops=1)" " Sort Key: (sum(t.price_amount)) DESC" " Sort Method: top-N heapsort Memory: 25kB" " -> HashAggregate (cost=354.64..357.14 rows=200 width=48) (actual time=44.267..44.289 rows=40 loops=1)" " Group Key: m.title" " Batches: 1 Memory Usage: 64kB" " -> Hash Join (cost=64.85..337.98 rows=3333 width=21) (actual time=1.220..41.143 rows=10000 loops=1)" " Hash Cond: (s.movie_id = m.id)" " -> Hash Join (cost=58.35..322.54 rows=3333 width=13) (actual time=0.774..38.604 rows=10000 loops=1)" " Hash Cond: (t.screening_id = s.id)" " -> Hash Join (cost=24.95..280.36 rows=3333 width=13) (actual time=0.720..36.364 rows=10000 loops=1)" " Hash Cond: (t.order_id = o.id)"	rows=200 width=48) (actual time=15563.512..15575.573 rows=3 loops=1)" " Sort Key: (sum(t.price_amount)) DESC" " Sort Method: top-N heapsort Memory: 25kB" " -> Finalize GroupAggregate (cost=212919.37..212971.54 rows=200 width=48) (actual time=15562.293..15575.357 rows=200 loops=1)" " Group Key: m.title" " -> Gather Merge (cost=212919.37..212966.04 rows=400 width=48) (actual time=15562.239..15574.593 rows=600 loops=1)" " Workers Planned: 2" " Workers Launched: 2" " -> Sort (cost=211919.35..211919.85 rows=200 width=48) (actual time=15472.005..15472.029 rows=200 loops=3)" " Sort Key: m.title" " Sort Method: quicksort Memory: 50kB" " Worker 0: Sort Method: quicksort Memory: 49kB" " Worker 1: Sort Method: quicksort Memory:		rows=200 width=48) (actual time=14692.569..14731.576 rows=3 loops=1)" " Sort Key: (sum(t.price_amount)) DESC" " Sort Method: top-N heapsort Memory: 25kB" " -> Finalize GroupAggregate (cost=212919.92..212972.09 rows=200 width=48) (actual time=14691.834..14731.436 rows=200 loops=1)" " Group Key: m.title" " -> Gather Merge (cost=212919.92..212966.59 rows=400 width=48) (actual time=14691.794..14730.974 rows=600 loops=1)" " Workers Planned: 2" " Workers Launched: 2" " -> Sort (cost=211919.90..211920.40 rows=200 width=48) (actual time=14539.176..14539.197 rows=200 loops=3)" " Sort Key: m.title" " Sort Method: quicksort Memory: 49kB" " Worker 0: Sort Method: quicksort Memory: 50kB" " Worker 1: Sort Method: quicksort Memory: 50kB"
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	" -> Seq Scan on tickets t (cost=0.00..229.00 rows=10000 width=21) (actual time=0.371..33.812 rows=10000 loops=1)" " Filter: ((status)::text = 'paid'::text)" " -> Hash (cost=22.08..22.08 rows=230 width=8) (actual time=0.328..0.329 rows=40 loops=1)" " Buckets: 1024 Batches: 1 Memory Usage: 10kB" " -> Seq Scan on orders o (cost=0.00..22.08 rows=230 width=8) (actual time=0.295..0.305 rows=40 loops=1)" " Filter: (paid_at >= (now() - '7 days'::interval))" " -> Hash (cost=20.40..20.40 rows=1040 width=16) (actual time=0.032..0.032 rows=40 loops=1)" " Buckets: 2048 Batches: 1 Memory Usage: 18kB" " -> Seq Scan on screenings s	50kB" " -> Partial HashAggregate (cost=211909.20..211911.70 rows=200 width=48) (actual time=15471.446..15471.553 rows=200 loops=3)" " Group Key: m.title" " Batches: 1 Memory Usage: 160kB" " Worker 0: Batches: 1 Memory Usage: 160kB" " Worker 1: Batches: 1 Memory Usage: 160kB" " -> Hash Join (cost=2853.50..191075.98 rows=4166645 width=21) (actual time=201.338..13960.030 rows=3333333 loops=3)" " Hash Cond: (t.order_id = o.id)" " -> Hash Join (cost=1319.50..178603.38 rows=4166645 width=29) (actual time=66.817..12969.465 rows=3333333 loops=3)" " Hash Cond: (s.movie_id = m.id)" " -> Hash Join (cost=1313.00..167427.67 rows=4166645 width=21) (actual		" -> Partial HashAggregate (cost=211909.75..211912.25 rows=200 width=48) (actual time=14538.703..14538.803 rows=200 loops=3)" " Group Key: m.title" " Batches: 1 Memory Usage: 160kB" " Worker 0: Batches: 1 Memory Usage: 160kB" " Worker 1: Batches: 1 Memory Usage: 160kB" " -> Hash Join (cost=2853.50..191076.42 rows=4166667 width=21) (actual time=102.799..13101.641 rows=3333333 loops=3)" " Hash Cond: (t.order_id = o.id)" " -> Hash Join (cost=1319.50..178603.76 rows=4166667 width=29) (actual time=27.766..12214.054 rows=3333333 loops=3)" " Hash Cond: (s.movie_id = m.id)" " -> Hash Join (cost=1313.00..167427.99 rows=4166667 width=21) (actual time=27.056..11373.819
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	(cost=0.00..20.40 rows=1040 width=16) (actual time=0.007..0.011 rows=40 loops=1)" " -> Hash (cost=4.00..4.00 rows=200 width=24) (actual time=0.418..0.419 rows=200 loops=1)" " Buckets: 1024 Batches: 1 Memory Usage: 19kB" " -> Seq Scan on movies m (cost=0.00..4.00 rows=200 width=24) (actual time=0.012..0.357 rows=200 loops=1)" "Planning Time: 17.769 ms" "Execution Time: 44.646 ms"	time=53.613..12081.178 rows=3333333 loops=3)" " Hash Cond: (t.screening_id = s.id)" " -> Parallel Seq Scan on tickets t (cost=0.00..155176.07 rows=4166645 width=21) (actual time=19.905..10932.363 rows=3333333 loops=3)" " Filter: ((status)::text = 'paid'::text)" " -> Hash (cost=813.00..813.00 rows=40000 width=16) (actual time=30.104..30.105 rows=40000 loops=3)" " Buckets: 65536 Batches: 1 Memory Usage: 2387kB" " - > Seq Scan on screenings s (cost=0.00..813.00 rows=40000 width=16) (actual time=2.559..11.067 rows=40000 loops=3)" " -> Hash (cost=4.00..4.00 rows=200 width=24) (actual time=13.133..13.134 rows=200 loops=3)" "	rows=3333333 loops=3)" " Hash Cond: (t.screening_id = s.id)" " -> Parallel Seq Scan on tickets t (cost=0.00..155176.33 rows=4166667 width=21) (actual time=0.766..10270.215 rows=3333333 loops=3)" " Filter: ((status)::text = 'paid'::text)" " -> Hash (cost=813.00..813.00 rows=40000 width=16) (actual time=25.458..25.458 rows=40000 loops=3)" " Buckets: 65536 Batches: 1 Memory Usage: 2387kB" " -> Seq Scan on screenings s (cost=0.00..813.00 rows=40000 width=16) (actual time=0.401..9.219 rows=40000 loops=3)" " -> Hash (cost=4.00..4.00 rows=200 width=24) (actual time=0.633..0.634 rows=200 loops=3)" " Buckets: 1024 Batches: 1 Memory Usage: 19kB" " ->
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		Buckets: 1024 Batches: 1 Memory Usage: 19kB" " -> Seq Scan on movies m (cost=0.00..4.00 rows=200 width=24) (actual time=12.979..13.017 rows=200 loops=3)" " -> Hash (cost=1034.00..1034.00 rows=40000 width=8) (actual time=133.678..133.679 rows=40000 loops=3)" " Buckets: 65536 Batches: 1 Memory Usage: 2075kB" " -> Seq Scan on orders o (cost=0.00..1034.00 rows=40000 width=8) (actual time=29.635..98.362 rows=40000 loops=3)" " Filter: (paid_at >= (now() - '7 days'::interval))" "Planning Time: 11.727 ms" "JIT:" " Functions: 106" " Options: Inlining false, Optimization false, Expressions true, Deforming true" " Timing: Generation 19.995 ms, Inlining 0.000 ms, Optimization 5.277 ms, Emission 142.151 ms,		Seq Scan on movies m (cost=0.00..4.00 rows=200 width=24) (actual time=0.471..0.508 rows=200 loops=3)" " -> Hash (cost=1034.00..1034.00 rows=40000 width=8) (actual time=74.052..74.053 rows=40000 loops=3)" " Buckets: 65536 Batches: 1 Memory Usage: 2075kB" " -> Seq Scan on orders o (cost=0.00..1034.00 rows=40000 width=8) (actual time=29.823..51.091 rows=40000 loops=3)" " Filter: (paid_at >= (now() - '7 days'::interval))" "Planning Time: 17.690 ms" "JIT:" " Functions: 106" " Options: Inlining false, Optimization false, Expressions true, Deforming true" " Timing: Generation 16.258 ms, Inlining 0.000 ms, Optimization 6.551 ms, Emission 176.220 ms, Total 199.029 ms" "Execution Time: 14831.533 ms"
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		Total 167.423 ms"		
		"Execution Time: 15644.925 ms"		
Сформировать схему зала и показать на ней свободные и занятые места на конкретный сеанс <pre> SELECT st.row_code, st.seat_number, CASE WHEN t.id IS NOT NULL THEN 'Занято' ELSE 'Свободно' END AS seat_status FROM seats st JOIN screenings s ON st.hall_id = s.hall_id LEFT JOIN tickets t ON st.id = t.seat_id AND t.screening_id = s.id AND t.status IN ('paid', 'reserved') </pre>	<pre> "Hash Left Join (cost=409.50..4722.50 rows=260000 width=38) (actual time=12.705..19.367 rows=10000 loops=1)" " Hash Cond: ((st.id = t.seat_id) AND (s.id = t.screening_id))" " -> Hash Join (cost=30.50..2978.50 rows=260000 width=22) (actual time=2.416..4.637 rows=10000 loops=1)" " Hash Cond: (s.hall_id = st.hall_id)" " -> Seq Scan on screenings s (cost=0.00..20.40 rows=1040 width=16) (actual time=0.022..0.045 rows=40 loops=1)" " -> Hash (cost=18.00..18.00 rows=1000 width=22) (actual time=2.369..2.370 rows=1000 loops=1)" " Buckets: 1024 Batches: 1 Memory Usage: 63kB" " -> Seq Scan on seats st (cost=0.00..18.00 rows=1000 width=22) (actual time=0.008..1.944 rows=1000 loops=1)" </pre>	<pre> "Hash Left Join (cost=436716.10..778411.10 rows=10000000 width=38) (actual time=40897.768..87954.347 rows=10000000 loops=1)" " Hash Cond: ((st.id = t.seat_id) AND (s.id = t.screening_id))" " -> Hash Join (cost=30.50..113443.50 rows=10000000 width=22) (actual time=679.408..1668.350 rows=10000000 loops=1)" " Hash Cond: (s.hall_id = st.hall_id)" " -> Seq Scan on screenings s (cost=0.00..813.00 rows=40000 width=16) (actual time=0.013..24.785 rows=40000 loops=1)" " -> Hash (cost=18.00..18.00 rows=1000 width=22) (actual time=679.345..679.346 rows=1000 loops=1)" " Buckets: 1024 Batches: 1 Memory Usage: 63kB" " -> Seq Scan on seats st (cost=0.00..18.00 rows=1000 width=22) (actual time=677.195..679.004 rows=1000 loops=1)" </pre>	Индексы <pre> CREATE INDEX idx_tickets_screening_seat_s tatus ON tickets (screening_id, seat_id) WHERE status IN ('paid', 'reserved'); CREATE INDEX idx_seats_hall_id ON seats (hall_id); </pre>	<pre> "Hash Left Join (cost=436717.50..778412.50 rows=10000000 width=38) (actual time=38706.095..85439.704 rows=10000000 loops=1)" " Hash Cond: ((st.id = t.seat_id) AND (s.id = t.screening_id))" " -> Hash Join (cost=30.50..113443.50 rows=10000000 width=22) (actual time=290.885..1383.067 rows=10000000 loops=1)" " Hash Cond: (s.hall_id = st.hall_id)" " -> Seq Scan on screenings s (cost=0.00..813.00 rows=40000 width=16) (actual time=0.021..27.474 rows=40000 loops=1)" " -> Hash (cost=18.00..18.00 rows=1000 width=22) (actual time=290.838..290.839 rows=1000 loops=1)" " Buckets: 1024 Batches: 1 Memory Usage: 63kB" " -> Seq Scan on seats st (cost=0.00..18.00 rows=1000 width=22) (actual time=290.498..290.607 rows=1000 loops=1)" " -> Hash </pre>

	" -> Hash (cost=229.00..229.00 rows=10000 width=24) (actual time=10.042..10.043 rows=10000 loops=1)" " Buckets: 16384 Batches: 1 Memory Usage: 675kB" " -> Seq Scan on tickets t (cost=0.00..229.00 rows=10000 width=24) (actual time=0.029..4.481 rows=10000 loops=1)" " Filter: ((status)::text = ANY ('{paid,reserved}'::text[]))" "Planning Time: 29.537 ms" "Execution Time: 20.345 ms"	" -> Hash (cost=228092.36..228092.36 rows=9999949 width=24) (actual time=39615.202..39615.203 rows=10000000 loops=1)" " Buckets: 131072 Batches: 128 Memory Usage: 5291kB" " -> Seq Scan on tickets t (cost=0.00..228092.36 rows=9999949 width=24) (actual time=0.307..24197.436 rows=10000000 loops=1)" " Filter: ((status)::text = ANY ('{paid,reserved}'::text[]))" "Planning Time: 8.126 ms" "JIT:" " Functions: 23" " Options: Inlining true, Optimization true, Expressions true, Deforming true" " Timing: Generation 2.046 ms, Inlining 237.642 ms, Optimization 289.256 ms, Emission 150.446 ms, Total 679.390 ms" "Execution Time: 88354.652 ms"		(cost=228093.00..228093.00 rows=10000000 width=24) (actual time=38302.957..38302.957 rows=10000000 loops=1)" " Buckets: 131072 Batches: 128 Memory Usage: 5291kB" " -> Seq Scan on tickets t (cost=0.00..228093.00 rows=10000000 width=24) (actual time=0.504..23219.086 rows=10000000 loops=1)" " Filter: ((status)::text = ANY ('{paid,reserved}'::text[]))" "Planning Time: 13.107 ms" "JIT:" " Functions: 23" " Options: Inlining true, Optimization true, Expressions true, Deforming true" " Timing: Generation 2.596 ms, Inlining 7.886 ms, Optimization 176.111 ms, Emission 106.599 ms, Total 293.193 ms" "Execution Time: 85841.436 ms"
Вывести диапазон минимальной и максимальной цены за билет на конкретный сеанс SELECT MIN(price_amount)	"Aggregate (cost=2.52..2.52 rows=1 width=64) (actual time=0.300..0.301 rows=1 loops=1)" " -> Seq Scan on screening_prices (cost=0.00..2.50 rows=3 width=5) (actual	"Aggregate (cost=11.99..12.00 rows=1 width=64) (actual time=1.208..1.209 rows=1 loops=1)" " -> Index Scan using screening_prices_pkey on screening_prices (cost=0.42..11.97 rows=3	Индекс CREATE INDEX idx_tickets_profit ON tickets(screening_id, status) INCLUDE (price_amount);	"Aggregate (cost=11.99..12.00 rows=1 width=64) (actual time=0.314..0.315 rows=1 loops=1)" " -> Index Scan using screening_prices_pkey on screening_prices (cost=0.42..11.97 rows=3 width=5)

AS min_price, MAX(price_amount) AS max_price FROM screening_prices WHERE screening_id = 2;	time=0.282..0.289 rows=3 loops=1)" " Filter: (screening_id = 2)" " Rows Removed by Filter: 117" "Planning Time: 12.610 ms" "Execution Time: 0.326 ms"	width=5) (actual time=1.190..1.193 rows=3 loops=1)" " Index Cond: (screening_id = 2)" "Planning Time: 4.260 ms" "Execution Time: 1.259 ms"		(actual time=0.265..0.271 rows=3 loops=1)" " Index Cond: (screening_id = 2)" "Planning Time: 0.294 ms" "Execution Time: 0.400 ms"
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